

import library

```
In [ ]: import numpy as np
import pandas as pd
```

Import data using read_csv()

```
In [ ]: data=pd.read_csv('/content/matches .csv')
```

Import function and attributes

```
In [ ]: # Check the type of the variable 'data' (e.g., DataFrame, Series, List, etc.)
type(data)
```

```
Out [ ]: pandas.core.frame.DataFrame
def __init__(data=None, index: Axes | None=None, columns: Axes | None=None, dtype: Dtype | None=None, copy: bool | None=None) -> None
```

Two-dimensional, size-mutable, potentially heterogeneous tabular data.

Data structure also contains labeled axes (rows and columns).

Arithmetic operations align on both row and column labels. Can be thought of as a dict-like container for Series objects. The primary

```
In [ ]: # Display the contents of the 'data' variable (usually a DataFrame)
data
```

Out[]:

	id	season	city	date	team1	team2	toss_winner	toss_decision
0	1	2017	Hyderabad	2017-04-05	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	field
1	2	2017	Pune	2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	field
2	3	2017	Rajkot	2017-04-07	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders	field
3	4	2017	Indore	2017-04-08	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab	field
4	5	2017	Bangalore	2017-04-08	Royal Challengers Bangalore	Delhi Daredevils	Royal Challengers Bangalore	bat
...	...	...	...	...	...	...	...	...
631	632	2016	Raipur	2016-05-22	Delhi Daredevils	Royal Challengers Bangalore	Royal Challengers Bangalore	field
632	633	2016	Bangalore	2016-05-24	Gujarat Lions	Royal Challengers Bangalore	Royal Challengers Bangalore	field
633	634	2016	Delhi	2016-05-25	Sunrisers Hyderabad	Kolkata Knight Riders	Kolkata Knight Riders	field
634	635	2016	Delhi	2016-05-27	Gujarat Lions	Sunrisers Hyderabad	Sunrisers Hyderabad	field
635	636	2016	Bangalore	2016-05-29	Sunrisers Hyderabad	Royal Challengers Bangalore	Sunrisers Hyderabad	bat

636 rows × 18 columns



In []:

```
# Display the first 5 rows of the DataFrame to quickly preview the data
data.head()
```

Out[]:

	id	season	city	date	team1	team2	toss_winner	toss_decision	result
0	1	2017	Hyderabad	2017-04-05	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	field	no
1	2	2017	Pune	2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	field	no
2	3	2017	Rajkot	2017-04-07	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders	field	no
3	4	2017	Indore	2017-04-08	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab	field	no
4	5	2017	Bangalore	2017-04-08	Royal Challengers Bangalore	Delhi Daredevils	Royal Challengers Bangalore	bat	no

In []: *# Display the first 2 rows of the DataFrame to quickly preview the data*

```
data.head(2)
```

Out[]:

	id	season	city	date	team1	team2	toss_winner	toss_decision	result
0	1	2017	Hyderabad	2017-04-05	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	field	no
1	2	2017	Pune	2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	field	no

In []: *# Display the Last 5 rows of the DataFrame to check the end of the dataset*

```
data.tail()
```

Out[]:

	id	season	city	date	team1	team2	toss_winner	toss_decision
631	632	2016	Raipur	2016-05-22	Delhi Daredevils	Royal Challengers Bangalore	Royal Challengers Bangalore	field
632	633	2016	Bangalore	2016-05-24	Gujarat Lions	Royal Challengers Bangalore	Royal Challengers Bangalore	field
633	634	2016	Delhi	2016-05-25	Sunrisers Hyderabad	Kolkata Knight Riders	Kolkata Knight Riders	field
634	635	2016	Delhi	2016-05-27	Gujarat Lions	Sunrisers Hyderabad	Sunrisers Hyderabad	field
635	636	2016	Bangalore	2016-05-29	Sunrisers Hyderabad	Royal Challengers Bangalore	Sunrisers Hyderabad	bat



In []:

```
# Get the number of rows and columns in the DataFrame as a tuple (rows, columns)

data.shape
```

Out[]: (636, 18)

In []:

```
# Display a summary of the DataFrame, including column names, data types, and no

data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 636 entries, 0 to 635
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   id                     636 non-null   int64
1   season                 636 non-null   int64
2   city                   629 non-null   object
3   date                   636 non-null   object
4   team1                  636 non-null   object
5   team2                  636 non-null   object
6   toss_winner            636 non-null   object
7   toss_decision          636 non-null   object
8   result                 636 non-null   object
9   dl_applied             636 non-null   int64
10  winner                  633 non-null   object
11  win_by_runs             636 non-null   int64
12  win_by_wickets          636 non-null   int64
13  player_of_match        633 non-null   object
14  venue                   636 non-null   object
15  umpire1                 635 non-null   object
16  umpire2                 635 non-null   object
17  umpire3                 0 non-null     float64
dtypes: float64(1), int64(5), object(12)
memory usage: 89.6+ KB
```

```
In [ ]: # Generate descriptive statistics (count, mean, std, min, max, etc.) for numeric
data.describe()
```

```
Out[ ]:
```

	id	season	dl_applied	win_by_runs	win_by_wickets	umpire3
count	636.000000	636.000000	636.000000	636.000000	636.000000	0.0
mean	318.500000	2012.490566	0.025157	13.682390	3.372642	NaN
std	183.741666	2.773026	0.156726	23.908877	3.420338	NaN
min	1.000000	2008.000000	0.000000	0.000000	0.000000	NaN
25%	159.750000	2010.000000	0.000000	0.000000	0.000000	NaN
50%	318.500000	2012.000000	0.000000	0.000000	4.000000	NaN
75%	477.250000	2015.000000	0.000000	20.000000	7.000000	NaN
max	636.000000	2017.000000	1.000000	146.000000	10.000000	NaN

Fetching rows and columns using iLoc

```
In [ ]: # Access the 'winner' column from the DataFrame (returns a Series)
data['winner']
```

```
Out[ ]:
```

	winner
0	Sunrisers Hyderabad
1	Rising Pune Supergiant
2	Kolkata Knight Riders
3	Kings XI Punjab
4	Royal Challengers Bangalore
...	...
631	Royal Challengers Bangalore
632	Royal Challengers Bangalore
633	Sunrisers Hyderabad
634	Sunrisers Hyderabad
635	Sunrisers Hyderabad

636 rows × 1 columns

dtype: object

```
In [ ]: type(data['winner']) # if i fetching 1 colomns or rows thats called series
```

Out[]: **pandas.core.series.Series**
 def __init__(data=None, index=None, dtype: Dtype | None=None, name=None, copy: bool | None=None, fastpath: bool | lib.NoDefault=lib.no_default) -> None

One-dimensional ndarray with axis labels (including time series).

Labels need not be unique but must be a hashable type. The object supports both integer- and label-based indexing and provides a host of methods for performing operations involving the index. Statistical

Action	Result Type
<code>data['column']</code>	Series (1D)
<code>data[['column']]</code>	DataFrame (2D)
<code>data.iloc[0]</code>	Series (1D row)
<code>data.iloc[[0]]</code>	DataFrame (1 row)
<code>data[['col1', 'col2']]</code>	DataFrame (2 cols)

In []: *# Select and display multiple columns ('team1', 'team2', and 'winner') from the*
`data[['team1', 'team2', 'winner']]`

Out[]:

	team1	team2	winner
0	Sunrisers Hyderabad	Royal Challengers Bangalore	Sunrisers Hyderabad
1	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant
2	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders
3	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab
4	Royal Challengers Bangalore	Delhi Daredevils	Royal Challengers Bangalore
...	...	...	...
631	Delhi Daredevils	Royal Challengers Bangalore	Royal Challengers Bangalore
632	Gujarat Lions	Royal Challengers Bangalore	Royal Challengers Bangalore
633	Sunrisers Hyderabad	Kolkata Knight Riders	Sunrisers Hyderabad
634	Gujarat Lions	Sunrisers Hyderabad	Sunrisers Hyderabad
635	Sunrisers Hyderabad	Royal Challengers Bangalore	Sunrisers Hyderabad

636 rows × 3 columns

```
In [ ]: # Select only the 'team1', 'team2', and 'winner' columns from the DataFrame
# and return the shape (number of rows and columns) of the resulting DataFrame

data[['team1', 'team2', 'winner']].shape
```

```
Out[ ]: (636, 3)
```

```
In [ ]: # Fetch the first row (index 0) of the DataFrame using integer-based indexing

data.iloc[0]
```

```
Out[ ]:
```

	0
id	1
season	2017
city	Hyderabad
date	2017-04-05
team1	Sunrisers Hyderabad
team2	Royal Challengers Bangalore
toss_winner	Royal Challengers Bangalore
toss_decision	field
result	normal
dl_applied	0
winner	Sunrisers Hyderabad
win_by_runs	35
win_by_wickets	0
player_of_match	Yuvraj Singh
venue	Rajiv Gandhi International Stadium, Uppal
umpire1	AY Dandekar
umpire2	NJ Llong
umpire3	NaN

dtype: object

```
In [ ]: # Select rows from index 1 to 10 (excluding 11) with a step of 2 using integer-b

data.iloc[1:11:2]
```

Out[]:

	id	season	city	date	team1	team2	toss_winner	toss_decision	result
1	2	2017	Pune	2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	field	normal
3	4	2017	Indore	2017-04-08	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab	field	normal
5	6	2017	Hyderabad	2017-04-09	Gujarat Lions	Sunrisers Hyderabad	Sunrisers Hyderabad	field	normal
7	8	2017	Indore	2017-04-10	Royal Challengers Bangalore	Kings XI Punjab	Royal Challengers Bangalore	bat	normal
9	10	2017	Mumbai	2017-04-12	Sunrisers Hyderabad	Mumbai Indians	Mumbai Indians	field	normal

In []: *# Select specific rows at index positions 1, 5, and 6 using integer-based indexing*

```
data.iloc[[1, 5, 6]]
```

Out[]:

	id	season	city	date	team1	team2	toss_winner	toss_decision	result
1	2	2017	Pune	2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	field	normal
5	6	2017	Hyderabad	2017-04-09	Gujarat Lions	Sunrisers Hyderabad	Sunrisers Hyderabad	field	normal
6	7	2017	Mumbai	2017-04-09	Kolkata Knight Riders	Mumbai Indians	Mumbai Indians	field	normal

In []: *# Select all rows (:) but only columns at index positions 4, 5, and 10 using integer-based indexing*

```
data.iloc[:, [4, 5, 10]]
```


Out[]:

	team1	team2	winner
0	Sunrisers Hyderabad	Royal Challengers Bangalore	Sunrisers Hyderabad
1	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant
2	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders
3	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab
4	Royal Challengers Bangalore	Delhi Daredevils	Royal Challengers Bangalore
...	...	...	...
631	Delhi Daredevils	Royal Challengers Bangalore	Royal Challengers Bangalore
632	Gujarat Lions	Royal Challengers Bangalore	Royal Challengers Bangalore
633	Sunrisers Hyderabad	Kolkata Knight Riders	Sunrisers Hyderabad
634	Gujarat Lions	Sunrisers Hyderabad	Sunrisers Hyderabad
635	Sunrisers Hyderabad	Royal Challengers Bangalore	Sunrisers Hyderabad

636 rows × 3 columns

Filtering dataframe on a condition

```
In [ ]: # Check which rows in the 'city' column have the value 'Hydrabad'

data['city'] == 'Hydrabad'
```

Out[]:

	city
0	False
1	False
2	False
3	False
4	False
...	...
631	False
632	False
633	False
634	False
635	False

636 rows × 1 columns

dtype: bool

In []:

```
# Create a boolean mask where city is 'Hyderabad'
mask = data['city'] == 'Hyderabad'

# Use the mask to filter and return only the rows where the city is Hyderabad
data[mask]
```

Out[]:

	id	season	city	date	team1	team2	toss_winner	toss_decision
0	1	2017	Hyderabad	2017-04-05	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	field
5	6	2017	Hyderabad	2017-04-09	Gujarat Lions	Sunrisers Hyderabad	Sunrisers Hyderabad	field
18	19	2017	Hyderabad	2017-04-17	Sunrisers Hyderabad	Kings XI Punjab	Kings XI Punjab	field
20	21	2017	Hyderabad	2017-04-19	Sunrisers Hyderabad	Delhi Daredevils	Sunrisers Hyderabad	bat
35	36	2017	Hyderabad	2017-04-30	Sunrisers Hyderabad	Kolkata Knight Riders	Kolkata Knight Riders	field
42	43	2017	Hyderabad	2017-05-06	Rising Pune Supergiant	Sunrisers Hyderabad	Sunrisers Hyderabad	field
46	47	2017	Hyderabad	2017-05-08	Mumbai Indians	Sunrisers Hyderabad	Mumbai Indians	bat
58	59	2017	Hyderabad	2017-05-21	Mumbai Indians	Rising Pune Supergiant	Mumbai Indians	bat
65	66	2008	Hyderabad	2008-04-22	Deccan Chargers	Delhi Daredevils	Deccan Chargers	bat
67	68	2008	Hyderabad	2008-04-24	Deccan Chargers	Rajasthan Royals	Rajasthan Royals	field
76	77	2008	Hyderabad	2008-05-01	Deccan Chargers	Kings XI Punjab	Kings XI Punjab	field
79	80	2008	Hyderabad	2008-05-25	Deccan Chargers	Royal Challengers Bangalore	Deccan Chargers	bat

	id	season	city	date	team1	team2	toss_winner	toss_decision
91	92	2008	Hyderabad	2008-05-11	Kolkata Knight Riders	Deccan Chargers	Kolkata Knight Riders	bat
101	102	2008	Hyderabad	2008-05-18	Mumbai Indians	Deccan Chargers	Deccan Chargers	field
113	114	2008	Hyderabad	2008-05-27	Deccan Chargers	Chennai Super Kings	Deccan Chargers	bat
235	236	2011	Hyderabad	2011-04-09	Deccan Chargers	Rajasthan Royals	Rajasthan Royals	field
244	245	2011	Hyderabad	2011-04-14	Deccan Chargers	Royal Challengers Bangalore	Royal Challengers Bangalore	field
248	249	2011	Hyderabad	2011-04-16	Deccan Chargers	Kings XI Punjab	Kings XI Punjab	field
259	260	2011	Hyderabad	2011-04-24	Mumbai Indians	Deccan Chargers	Deccan Chargers	field
274	275	2011	Hyderabad	2011-05-03	Kolkata Knight Riders	Deccan Chargers	Deccan Chargers	field
278	279	2011	Hyderabad	2011-05-05	Deccan Chargers	Delhi Daredevils	Delhi Daredevils	field
285	286	2011	Hyderabad	2011-05-10	Deccan Chargers	Pune Warriors	Deccan Chargers	bat
330	331	2012	Hyderabad	2012-05-10	Deccan Chargers	Delhi Daredevils	Deccan Chargers	bat
358	359	2012	Hyderabad	2012-05-08	Kings XI Punjab	Deccan Chargers	Deccan Chargers	field

	id	season	city	date	team1	team2	toss_winner	toss_decision
372	373	2012	Hyderabad	2012-05-18	Rajasthan Royals	Deccan Chargers	Rajasthan Royals	bat
375	376	2012	Hyderabad	2012-05-20	Deccan Chargers	Royal Challengers Bangalore	Royal Challengers Bangalore	field
383	384	2013	Hyderabad	2013-04-05	Sunrisers Hyderabad	Pune Warriors	Pune Warriors	field
387	388	2013	Hyderabad	2013-04-07	Royal Challengers Bangalore	Sunrisers Hyderabad	Royal Challengers Bangalore	bat
404	405	2013	Hyderabad	2013-04-19	Kings XI Punjab	Sunrisers Hyderabad	Kings XI Punjab	bat
422	423	2013	Hyderabad	2013-05-01	Mumbai Indians	Sunrisers Hyderabad	Mumbai Indians	bat
427	428	2013	Hyderabad	2013-05-04	Delhi Daredevils	Sunrisers Hyderabad	Delhi Daredevils	bat
434	435	2013	Hyderabad	2013-05-08	Chennai Super Kings	Sunrisers Hyderabad	Sunrisers Hyderabad	field
448	449	2013	Hyderabad	2013-05-17	Sunrisers Hyderabad	Rajasthan Royals	Sunrisers Hyderabad	bat
452	453	2013	Hyderabad	2013-05-19	Kolkata Knight Riders	Sunrisers Hyderabad	Kolkata Knight Riders	bat
492	493	2014	Hyderabad	2014-05-12	Sunrisers Hyderabad	Mumbai Indians	Sunrisers Hyderabad	bat
495	496	2014	Hyderabad	2014-05-14	Sunrisers Hyderabad	Kings XI Punjab	Kings XI Punjab	field

	id	season	city	date	team1	team2	toss_winner	toss_decision
499	500	2014	Hyderabad	2014-05-18	Sunrisers Hyderabad	Kolkata Knight Riders	Sunrisers Hyderabad	bat
502	503	2014	Hyderabad	2014-05-20	Royal Challengers Bangalore	Sunrisers Hyderabad	Royal Challengers Bangalore	bat
550	551	2015	Hyderabad	2015-05-02	Sunrisers Hyderabad	Chennai Super Kings	Chennai Super Kings	field
563	564	2015	Hyderabad	2015-05-11	Sunrisers Hyderabad	Kings XI Punjab	Sunrisers Hyderabad	bat
567	568	2015	Hyderabad	2015-05-15	Sunrisers Hyderabad	Royal Challengers Bangalore	Sunrisers Hyderabad	bat
571	572	2015	Hyderabad	2015-05-17	Sunrisers Hyderabad	Mumbai Indians	Sunrisers Hyderabad	bat
583	584	2016	Hyderabad	2016-04-16	Sunrisers Hyderabad	Kolkata Knight Riders	Sunrisers Hyderabad	bat
587	588	2016	Hyderabad	2016-04-18	Mumbai Indians	Sunrisers Hyderabad	Sunrisers Hyderabad	field
593	594	2016	Hyderabad	2016-04-23	Kings XI Punjab	Sunrisers Hyderabad	Sunrisers Hyderabad	field
597	598	2016	Hyderabad	2016-04-26	Sunrisers Hyderabad	Rising Pune Supergiants	Rising Pune Supergiants	field
602	603	2016	Hyderabad	2016-04-30	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	field
609	610	2016	Hyderabad	2016-05-06	Gujarat Lions	Sunrisers Hyderabad	Sunrisers Hyderabad	field

	id	season	city	date	team1	team2	toss_winner	toss_decision
617	618	2016	Hyderabad	2016-05-12	Sunrisers Hyderabad	Delhi Daredevils	Delhi Daredevils	field

```
In [ ]: # Create a mask for rows where the city is 'Hyderabad'
mask = data['city'] == 'Hyderabad'

# Apply the mask and check how many rows and columns are in the filtered data
data[mask].shape
```

Out[]: (49, 18)

```
In [ ]: # Create a mask to filter rows where the city is 'Hyderabad'
mask = data['city'] == 'Hyderabad'

# Apply the mask and get only the number of rows (i.e., matches in Hyderabad)
data[mask].shape[0]
```

Out[]: 49

```
In [ ]: # Define a function that takes a city name as input
def get_city_matches_count(city):
    # Create a boolean mask where the 'city' column matches the input city
    mask = data['city'] == city

    # Use the mask to filter the DataFrame and return only the number of matches
    return data[mask].shape[0]
```

```
In [ ]: # Call the function to get the number of matches played in the city 'Bangalore'
get_city_matches_count('Bangalore')
```

Out[]: 66

```
In [ ]: # Create a boolean mask to filter rows where the 'city' is 'Hyderabad'
mask1 = data['city'] == 'Hyderabad'

# Create another boolean mask to filter rows where the 'date' is after January 1, 2010
mask2 = data['date'] > '2010-01-01'

# Apply both conditions using the '&' (AND) operator
# This returns only those rows where both conditions are True
data[mask1 & mask2]
```

Out[]:

	id	season	city	date	team1	team2	toss_winner	toss_decision
0	1	2017	Hyderabad	2017-04-05	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	field
5	6	2017	Hyderabad	2017-04-09	Gujarat Lions	Sunrisers Hyderabad	Sunrisers Hyderabad	field
18	19	2017	Hyderabad	2017-04-17	Sunrisers Hyderabad	Kings XI Punjab	Kings XI Punjab	field
20	21	2017	Hyderabad	2017-04-19	Sunrisers Hyderabad	Delhi Daredevils	Sunrisers Hyderabad	bat
35	36	2017	Hyderabad	2017-04-30	Sunrisers Hyderabad	Kolkata Knight Riders	Kolkata Knight Riders	field
42	43	2017	Hyderabad	2017-05-06	Rising Pune Supergiant	Sunrisers Hyderabad	Sunrisers Hyderabad	field
46	47	2017	Hyderabad	2017-05-08	Mumbai Indians	Sunrisers Hyderabad	Mumbai Indians	bat
58	59	2017	Hyderabad	2017-05-21	Mumbai Indians	Rising Pune Supergiant	Mumbai Indians	bat
235	236	2011	Hyderabad	2011-04-09	Deccan Chargers	Rajasthan Royals	Rajasthan Royals	field
244	245	2011	Hyderabad	2011-04-14	Deccan Chargers	Royal Challengers Bangalore	Royal Challengers Bangalore	field
248	249	2011	Hyderabad	2011-04-16	Deccan Chargers	Kings XI Punjab	Kings XI Punjab	field
259	260	2011	Hyderabad	2011-04-24	Mumbai Indians	Deccan Chargers	Deccan Chargers	field

	id	season	city	date	team1	team2	toss_winner	toss_decision
274	275	2011	Hyderabad	2011-05-03	Kolkata Knight Riders	Deccan Chargers	Deccan Chargers	field
278	279	2011	Hyderabad	2011-05-05	Deccan Chargers	Delhi Daredevils	Delhi Daredevils	field
285	286	2011	Hyderabad	2011-05-10	Deccan Chargers	Pune Warriors	Deccan Chargers	bat
330	331	2012	Hyderabad	2012-05-10	Deccan Chargers	Delhi Daredevils	Deccan Chargers	bat
358	359	2012	Hyderabad	2012-05-08	Kings XI Punjab	Deccan Chargers	Deccan Chargers	field
372	373	2012	Hyderabad	2012-05-18	Rajasthan Royals	Deccan Chargers	Rajasthan Royals	bat
375	376	2012	Hyderabad	2012-05-20	Deccan Chargers	Royal Challengers Bangalore	Royal Challengers Bangalore	field
383	384	2013	Hyderabad	2013-04-05	Sunrisers Hyderabad	Pune Warriors	Pune Warriors	field
387	388	2013	Hyderabad	2013-04-07	Royal Challengers Bangalore	Sunrisers Hyderabad	Royal Challengers Bangalore	bat
404	405	2013	Hyderabad	2013-04-19	Kings XI Punjab	Sunrisers Hyderabad	Kings XI Punjab	bat
422	423	2013	Hyderabad	2013-05-01	Mumbai Indians	Sunrisers Hyderabad	Mumbai Indians	bat
427	428	2013	Hyderabad	2013-05-04	Delhi Daredevils	Sunrisers Hyderabad	Delhi Daredevils	bat

	id	season	city	date	team1	team2	toss_winner	toss_decision
434	435	2013	Hyderabad	2013-05-08	Chennai Super Kings	Sunrisers Hyderabad	Sunrisers Hyderabad	field
448	449	2013	Hyderabad	2013-05-17	Sunrisers Hyderabad	Rajasthan Royals	Sunrisers Hyderabad	bat
452	453	2013	Hyderabad	2013-05-19	Kolkata Knight Riders	Sunrisers Hyderabad	Kolkata Knight Riders	bat
492	493	2014	Hyderabad	2014-05-12	Sunrisers Hyderabad	Mumbai Indians	Sunrisers Hyderabad	bat
495	496	2014	Hyderabad	2014-05-14	Sunrisers Hyderabad	Kings XI Punjab	Kings XI Punjab	field
499	500	2014	Hyderabad	2014-05-18	Sunrisers Hyderabad	Kolkata Knight Riders	Sunrisers Hyderabad	bat
502	503	2014	Hyderabad	2014-05-20	Royal Challengers Bangalore	Sunrisers Hyderabad	Royal Challengers Bangalore	bat
550	551	2015	Hyderabad	2015-05-02	Sunrisers Hyderabad	Chennai Super Kings	Chennai Super Kings	field
563	564	2015	Hyderabad	2015-05-11	Sunrisers Hyderabad	Kings XI Punjab	Sunrisers Hyderabad	bat
567	568	2015	Hyderabad	2015-05-15	Sunrisers Hyderabad	Royal Challengers Bangalore	Sunrisers Hyderabad	bat
571	572	2015	Hyderabad	2015-05-17	Sunrisers Hyderabad	Mumbai Indians	Sunrisers Hyderabad	bat
583	584	2016	Hyderabad	2016-04-16	Sunrisers Hyderabad	Kolkata Knight Riders	Sunrisers Hyderabad	bat

	id	season	city	date	team1	team2	toss_winner	toss_decision
587	588	2016	Hyderabad	2016-04-18	Mumbai Indians	Sunrisers Hyderabad	Sunrisers Hyderabad	field
593	594	2016	Hyderabad	2016-04-23	Kings XI Punjab	Sunrisers Hyderabad	Sunrisers Hyderabad	field
597	598	2016	Hyderabad	2016-04-26	Sunrisers Hyderabad	Rising Pune Supergiants	Rising Pune Supergiants	field
602	603	2016	Hyderabad	2016-04-30	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	field
609	610	2016	Hyderabad	2016-05-06	Gujarat Lions	Sunrisers Hyderabad	Sunrisers Hyderabad	field
617	618	2016	Hyderabad	2016-05-12	Sunrisers Hyderabad	Delhi Daredevils	Delhi Daredevils	field

```
In [ ]: mask1=data['city']=='Hyderabad'
mask2=data['date']>'2010-01-01'

# Apply both conditions using logical AND (&), then count matching rows
data[mask1 & mask2].shape[0]
```

Out[]: 42

The value_counts() function

```
In [ ]: # Select the 'winner' column and count how many times each team appears as a win
# This helps identify which teams won the most matches

data['winner'].value_counts()
```

Out[]:

	count
winner	
Mumbai Indians	92
Chennai Super Kings	79
Kolkata Knight Riders	77
Royal Challengers Bangalore	73
Kings XI Punjab	70
Rajasthan Royals	63
Delhi Daredevils	62
Sunrisers Hyderabad	42
Deccan Chargers	29
Gujarat Lions	13
Pune Warriors	12
Rising Pune Supergiant	10
Kochi Tuskers Kerala	6
Rising Pune Supergiants	5

dtype: int64

The Plot function

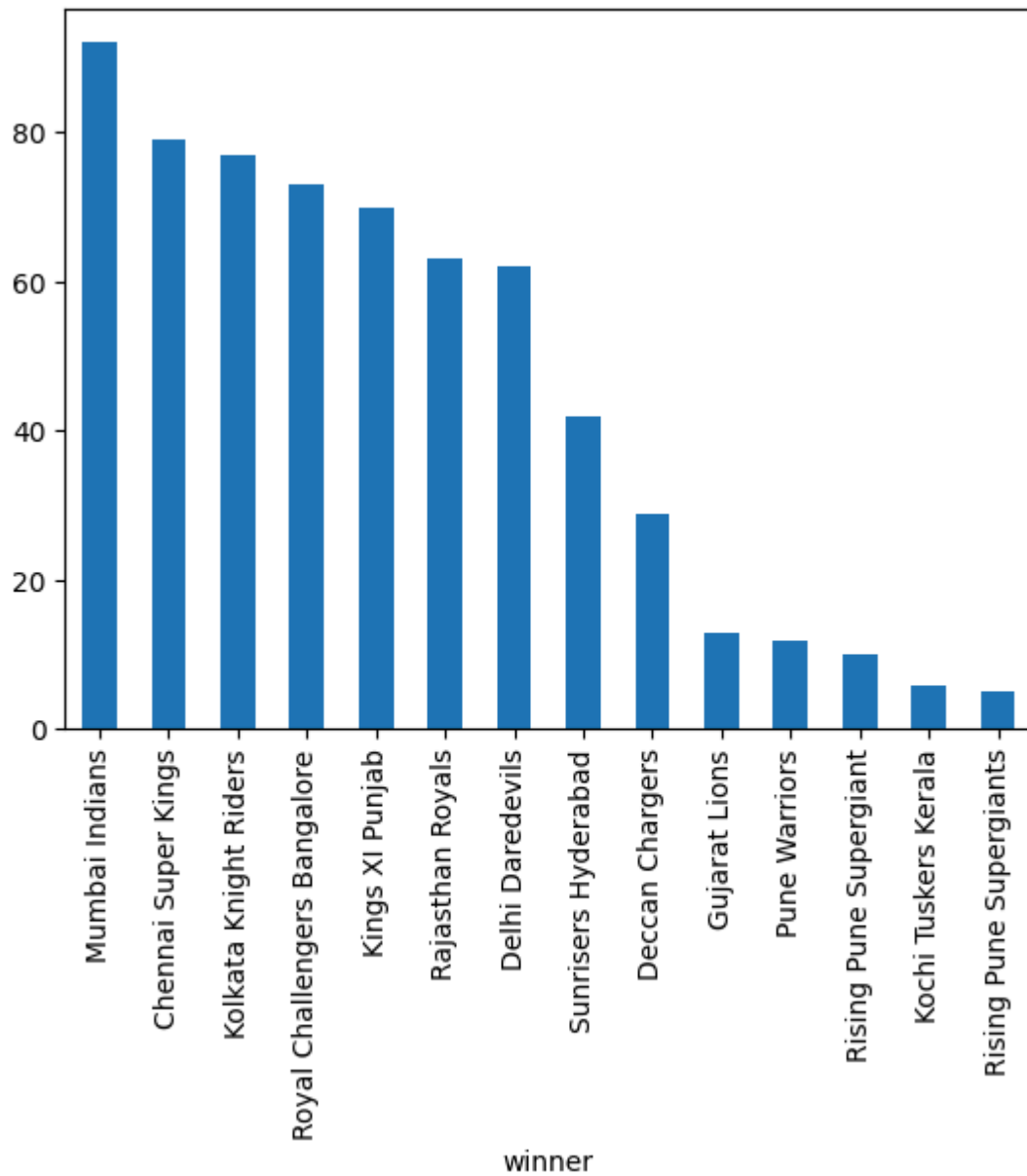
```
In [ ]: # Importing the matplotlib library for data visualization
# 'plt' is the standard alias used for creating plots and charts

import matplotlib.pyplot as plt
```

```
In [ ]: # Plot a bar chart of the number of wins by each team
# 'value_counts()' counts how many times each team has won
# 'plot(kind="bar")' creates a vertical bar chart

data['winner'].value_counts().plot(kind='bar')
```

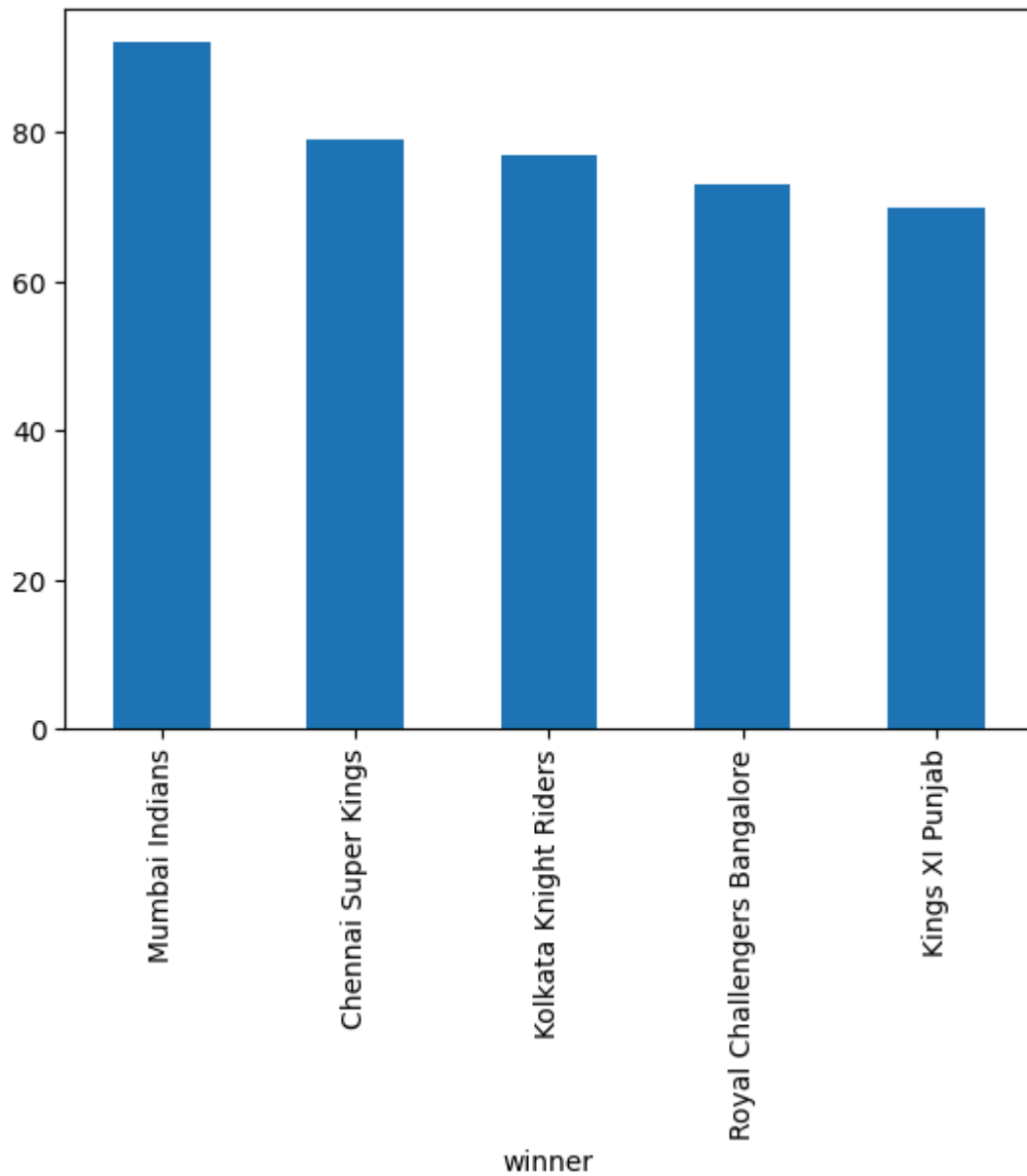
Out[]: <Axes: xlabel='winner'>



```
In [ ]: # Get the top 5 teams with the most wins and plot a bar chart
# 'value_counts()' counts the wins per team
# 'head()' selects the top 5 teams
# 'plot(kind="bar")' creates a vertical bar chart for those 5

data['winner'].value_counts().head().plot(kind='bar')
```

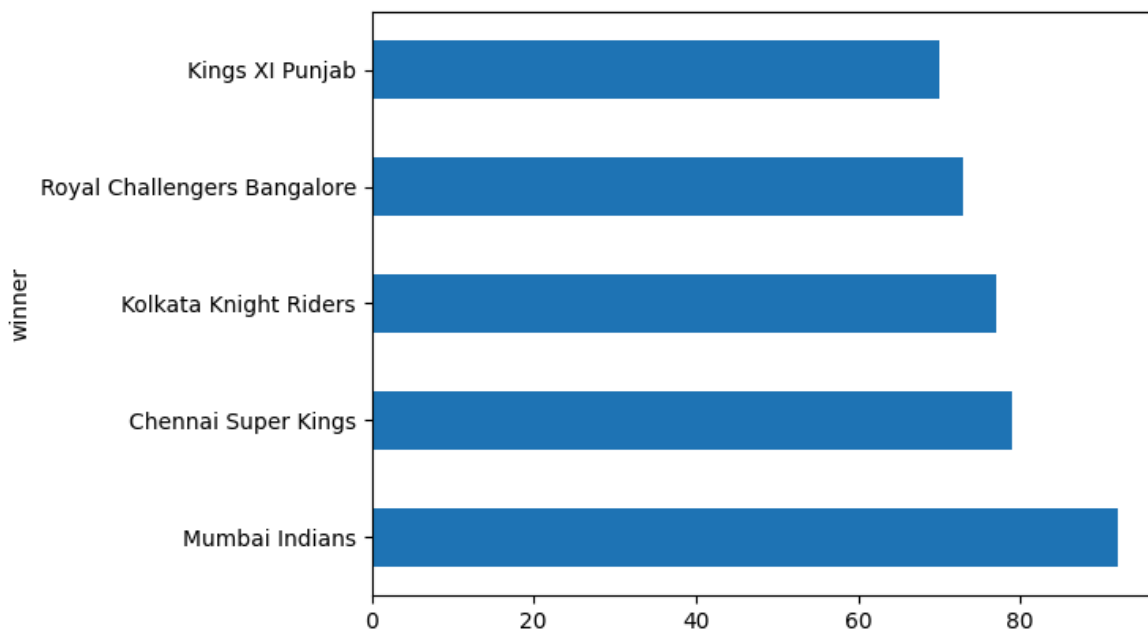
```
Out[ ]: <Axes: xlabel='winner'>
```



```
In [ ]: # Plot a horizontal bar chart of the top 5 most winning teams
# 'value_counts()' counts the number of wins per team
# 'head()' selects the top 5 teams
# 'plot(kind="barh")' plots a horizontal bar chart

data['winner'].value_counts().head().plot(kind='barh')
```

```
Out[ ]: <Axes: ylabel='winner'>
```



```
In [ ]: # Count how many times each toss decision was made (e.g., bat or field)

data['toss_decision'].value_counts()
```

```
Out[ ]:          count
```

```
  toss_decision
```

```
    field    363
```

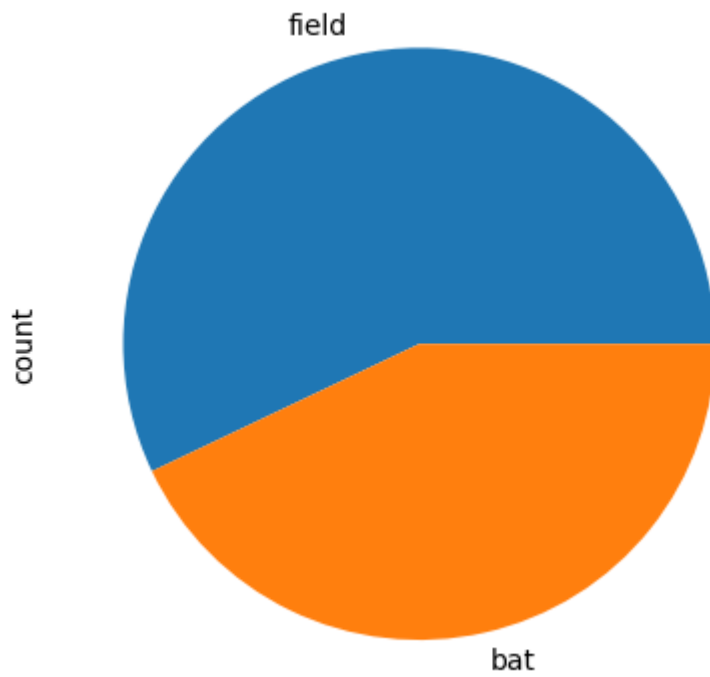
```
    bat     273
```

```
dtype: int64
```

```
In [ ]: # Plot a pie chart of toss decisions (bat vs field)
# value_counts() counts each decision
# plot(kind='pie') generates a pie chart

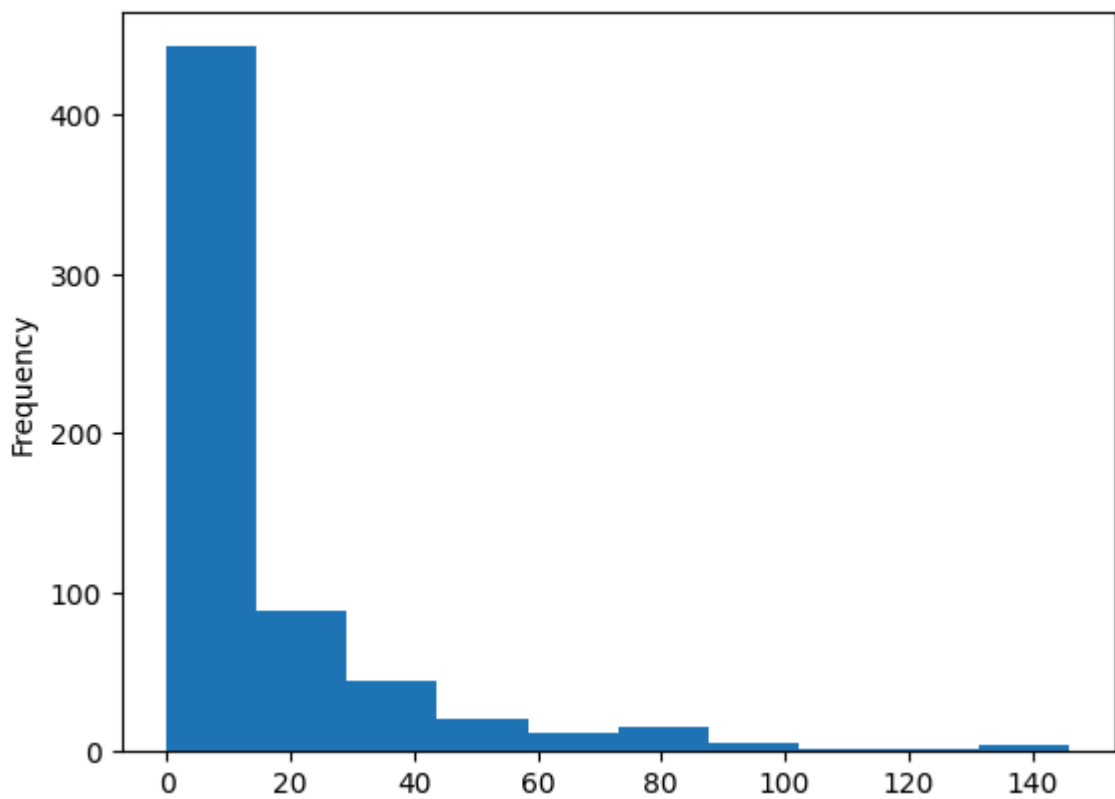
data['toss_decision'].value_counts().plot(kind='pie')
```

```
Out[ ]: <Axes: ylabel='count'>
```



```
In [ ]: # Plot a histogram to show the distribution of match victories by run margin  
data['win_by_runs'].plot(kind='hist')
```

```
Out[ ]: <Axes: ylabel='Frequency'>
```



Series Operations


```
In [ ]: # Create a Series containing the number of matches won by each team
```

```
myseries = data['winner'].value_counts()
```

```
In [ ]: # Display the index of the Series, i.e., the team names (winners)
```

```
myseries.index
```

```
Out[ ]: Index(['Mumbai Indians', 'Chennai Super Kings', 'Kolkata Knight Riders',
              'Royal Challengers Bangalore', 'Kings XI Punjab', 'Rajasthan Royals',
              'Delhi Daredevils', 'Sunrisers Hyderabad', 'Deccan Chargers',
              'Gujarat Lions', 'Pune Warriors', 'Rising Pune Supergiant',
              'Kochi Tuskers Kerala', 'Rising Pune Supergiants'],
              dtype='object', name='winner')
```

```
In [ ]: # Count how many teams have the same number of wins
```

```
myseries.value_counts()
```

```
Out[ ]:      count
```

```
count
```

```
92      1
```

```
79      1
```

```
77      1
```

```
73      1
```

```
70      1
```

```
63      1
```

```
62      1
```

```
42      1
```

```
29      1
```

```
13      1
```

```
12      1
```

```
10      1
```

```
6       1
```

```
5       1
```

```
dtype: int64
```

```
In [ ]: # Display the top 5 teams with the most match wins
```

```
myseries.head()
```

Out[]: **count**

winner	
Mumbai Indians	92
Chennai Super Kings	79
Kolkata Knight Riders	77
Royal Challengers Bangalore	73
Kings XI Punjab	70

dtype: int64

```
In [ ]: # Get the total number of matches won by Royal Challengers Bangalore
myseries['Royal Challengers Bangalore']
```

Out[]: np.int64(73)

```
In [ ]: # Total number of matches played by each team (as team1 or team2)
data['team1'].value_counts() + data['team2'].value_counts()
```

Out[]: **count**

Chennai Super Kings	131
Deccan Chargers	75
Delhi Daredevils	147
Gujarat Lions	30
Kings XI Punjab	148
Kochi Tuskers Kerala	14
Kolkata Knight Riders	148
Mumbai Indians	157
Pune Warriors	46
Rajasthan Royals	118
Rising Pune Supergiant	16
Rising Pune Supergiants	14
Royal Challengers Bangalore	152
Sunrisers Hyderabad	76

dtype: int64

The sort_values() method

```
In [ ]: # Sort the result in ascending order (least to most matches played)

(data['team1'].value_counts() + data['team2'].value_counts()).sort_values()
```

```
Out[ ]:
```

	count
Kochi Tuskers Kerala	14
Rising Pune Supergiants	14
Rising Pune Supergiant	16
Gujarat Lions	30
Pune Warriors	46
Deccan Chargers	75
Sunrisers Hyderabad	76
Rajasthan Royals	118
Chennai Super Kings	131
Delhi Daredevils	147
Kings XI Punjab	148
Kolkata Knight Riders	148
Royal Challengers Bangalore	152
Mumbai Indians	157

dtype: int64

```
In [ ]: # Sort the teams by total matches played in descending order (most to least)

(data['team1'].value_counts() + data['team2'].value_counts()).sort_values(ascending=False)
```

Out[]:

	count
Mumbai Indians	157
Royal Challengers Bangalore	152
Kings XI Punjab	148
Kolkata Knight Riders	148
Delhi Daredevils	147
Chennai Super Kings	131
Rajasthan Royals	118
Sunrisers Hyderabad	76
Deccan Chargers	75
Pune Warriors	46
Gujarat Lions	30
Rising Pune Supergiant	16
Kochi Tuskers Kerala	14
Rising Pune Supergiants	14

dtype: int64

In []:

data

Out[]:

	id	season	city	date	team1	team2	toss_winner	toss_decision
0	1	2017	Hyderabad	2017-04-05	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	field
1	2	2017	Pune	2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	field
2	3	2017	Rajkot	2017-04-07	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders	field
3	4	2017	Indore	2017-04-08	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab	field
4	5	2017	Bangalore	2017-04-08	Royal Challengers Bangalore	Delhi Daredevils	Royal Challengers Bangalore	bat
...	...	...	...	...	...	...	...	...
631	632	2016	Raipur	2016-05-22	Delhi Daredevils	Royal Challengers Bangalore	Royal Challengers Bangalore	field
632	633	2016	Bangalore	2016-05-24	Gujarat Lions	Royal Challengers Bangalore	Royal Challengers Bangalore	field
633	634	2016	Delhi	2016-05-25	Sunrisers Hyderabad	Kolkata Knight Riders	Kolkata Knight Riders	field
634	635	2016	Delhi	2016-05-27	Gujarat Lions	Sunrisers Hyderabad	Sunrisers Hyderabad	field
635	636	2016	Bangalore	2016-05-29	Sunrisers Hyderabad	Royal Challengers Bangalore	Sunrisers Hyderabad	bat

636 rows × 18 columns



In []:

```
# Sort the DataFrame by the 'city' column in ascending order
data.sort_values('city')
```

Out[]:

	id	season	city	date	team1	team2	toss_winner	toss_decision	res
460	461	2014	Abu Dhabi	2014-04-18	Sunrisers Hyderabad	Rajasthan Royals	Rajasthan Royals	field	nor
471	472	2014	Abu Dhabi	2014-04-26	Kings XI Punjab	Kolkata Knight Riders	Kolkata Knight Riders	field	nor
470	471	2014	Abu Dhabi	2014-04-26	Royal Challengers Bangalore	Rajasthan Royals	Rajasthan Royals	field	nor
464	465	2014	Abu Dhabi	2014-04-21	Chennai Super Kings	Delhi Daredevils	Chennai Super Kings	bat	nor
459	460	2014	Abu Dhabi	2014-04-18	Chennai Super Kings	Kings XI Punjab	Chennai Super Kings	bat	nor
...	...	...	...	...	...	...	...	...	...
466	467	2014	NaN	2014-04-23	Chennai Super Kings	Rajasthan Royals	Rajasthan Royals	field	nor
468	469	2014	NaN	2014-04-25	Sunrisers Hyderabad	Delhi Daredevils	Sunrisers Hyderabad	bat	nor
469	470	2014	NaN	2014-04-25	Mumbai Indians	Chennai Super Kings	Mumbai Indians	bat	nor
474	475	2014	NaN	2014-04-28	Royal Challengers Bangalore	Kings XI Punjab	Kings XI Punjab	field	nor
476	477	2014	NaN	2014-04-30	Sunrisers Hyderabad	Mumbai Indians	Mumbai Indians	field	nor

636 rows × 18 columns



In []:

```
# Sort the DataFrame by the 'city' column in descending order
# Set inplace=True to modify the original DataFrame instead of returning a new one
data.sort_values('city', ascending=False) #inplace=True
```

Out[]:

	id	season	city	date	team1	team2	toss_winner	toss_deci
312	313	2012	Visakhapatnam	2012-04-07	Chennai Super Kings	Deccan Chargers	Deccan Chargers	
622	623	2016	Visakhapatnam	2016-05-15	Mumbai Indians	Delhi Daredevils	Delhi Daredevils	
615	616	2016	Visakhapatnam	2016-05-10	Sunrisers Hyderabad	Rising Pune Supergiants	Sunrisers Hyderabad	
612	613	2016	Visakhapatnam	2016-05-08	Sunrisers Hyderabad	Mumbai Indians	Mumbai Indians	
624	625	2016	Visakhapatnam	2016-05-17	Delhi Daredevils	Rising Pune Supergiants	Rising Pune Supergiants	
...	...	...	...	...	...	...	...	...
466	467	2014	NaN	2014-04-23	Chennai Super Kings	Rajasthan Royals	Rajasthan Royals	
468	469	2014	NaN	2014-04-25	Sunrisers Hyderabad	Delhi Daredevils	Sunrisers Hyderabad	
469	470	2014	NaN	2014-04-25	Mumbai Indians	Chennai Super Kings	Mumbai Indians	
474	475	2014	NaN	2014-04-28	Royal Challengers Bangalore	Kings XI Punjab	Kings XI Punjab	
476	477	2014	NaN	2014-04-30	Sunrisers Hyderabad	Mumbai Indians	Mumbai Indians	

636 rows × 18 columns

```
In [ ]: # Sort the DataFrame first by 'city' in ascending order (A to Z),
# and then by 'date' in descending order (latest date first) within each city group
data.sort_values(['city', 'date'], ascending=[True, False])
```

```
Out[ ]:
```

	id	season	city	date	team1	team2	toss_winner	toss_decision	result
475	476	2014	Abu Dhabi	2014-04-29	Rajasthan Royals	Kolkata Knight Riders	Rajasthan Royals	bat	
470	471	2014	Abu Dhabi	2014-04-26	Royal Challengers Bangalore	Rajasthan Royals	Rajasthan Royals	field	no
471	472	2014	Abu Dhabi	2014-04-26	Kings XI Punjab	Kolkata Knight Riders	Kolkata Knight Riders	field	no
464	465	2014	Abu Dhabi	2014-04-21	Chennai Super Kings	Delhi Daredevils	Chennai Super Kings	bat	no
459	460	2014	Abu Dhabi	2014-04-18	Chennai Super Kings	Kings XI Punjab	Chennai Super Kings	bat	no
...	...	...	...	...	...	...	...	...	...
468	469	2014	NaN	2014-04-25	Sunrisers Hyderabad	Delhi Daredevils	Sunrisers Hyderabad	bat	no
469	470	2014	NaN	2014-04-25	Mumbai Indians	Chennai Super Kings	Mumbai Indians	bat	no
466	467	2014	NaN	2014-04-23	Chennai Super Kings	Rajasthan Royals	Rajasthan Royals	field	no
461	462	2014	NaN	2014-04-19	Mumbai Indians	Royal Challengers Bangalore	Royal Challengers Bangalore	field	no
462	463	2014	NaN	2014-04-19	Kolkata Knight Riders	Delhi Daredevils	Kolkata Knight Riders	bat	no

636 rows × 10 columns

The drop_duplicates() method

```
In [ ]: # Drop duplicate rows based on the 'city' column  
# Keeps the first occurrence of each unique city and removes the rest  
  
data.drop_duplicates(subset=['city'])
```

Out[]:

	id	season	city	date	team1	team2	toss_winner	toss_deci
0	1	2017	Hyderabad	2017-04-05	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	
1	2	2017	Pune	2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	
2	3	2017	Rajkot	2017-04-07	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders	
3	4	2017	Indore	2017-04-08	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab	
4	5	2017	Bangalore	2017-04-08	Royal Challengers Bangalore	Delhi Daredevils	Royal Challengers Bangalore	
6	7	2017	Mumbai	2017-04-09	Kolkata Knight Riders	Mumbai Indians	Mumbai Indians	
10	11	2017	Kolkata	2017-04-13	Kings XI Punjab	Kolkata Knight Riders	Kolkata Knight Riders	
14	15	2017	Delhi	2017-04-15	Delhi Daredevils	Kings XI Punjab	Delhi Daredevils	
31	32	2017	Chandigarh	2017-04-28	Sunrisers Hyderabad	Kings XI Punjab	Kings XI Punjab	
48	49	2017	Kanpur	2017-05-10	Gujarat Lions	Delhi Daredevils	Delhi Daredevils	
64	65	2008	Jaipur	2008-04-21	Kings XI Punjab	Rajasthan Royals	Kings XI Punjab	
66	67	2008	Chennai	2008-04-23	Chennai Super Kings	Mumbai Indians	Mumbai Indians	
117	118	2009	Cape Town	2009-04-18	Mumbai Indians	Chennai Super Kings	Chennai Super Kings	
121	122	2009	Port Elizabeth	2009-04-20	Chennai Super Kings	Royal Challengers Bangalore	Chennai Super Kings	
122	123	2009	Durban	2009-04-21	Kings XI Punjab	Kolkata Knight	Kolkata Knight	

	id	season	city	date	team1	team2	toss_winner	toss_deci
						Riders	Riders	
132	133	2009	Centurion	2009-04-28	Delhi Daredevils	Rajasthan Royals	Delhi Daredevils	
137	138	2009	East London	2009-05-01	Mumbai Indians	Kolkata Knight Riders	Mumbai Indians	
140	141	2009	Johannesburg	2009-05-02	Chennai Super Kings	Delhi Daredevils	Delhi Daredevils	
150	151	2009	Kimberley	2009-05-09	Deccan Chargers	Kings XI Punjab	Kings XI Punjab	
160	161	2009	Bloemfontein	2009-05-15	Delhi Daredevils	Kings XI Punjab	Kings XI Punjab	
179	180	2010	Ahmedabad	2010-03-15	Rajasthan Royals	Delhi Daredevils	Delhi Daredevils	
185	186	2010	Cuttack	2010-03-19	Deccan Chargers	Kings XI Punjab	Kings XI Punjab	
209	210	2010	Nagpur	2010-04-05	Rajasthan Royals	Deccan Chargers	Rajasthan Royals	
224	225	2010	Dharamsala	2010-04-16	Kings XI Punjab	Deccan Chargers	Deccan Chargers	
236	237	2011	Kochi	2011-04-09	Kochi Tuskers Kerala	Royal Challengers Bangalore	Kochi Tuskers Kerala	
312	313	2012	Visakhapatnam	2012-04-07	Chennai Super Kings	Deccan Chargers	Deccan Chargers	
418	419	2013	Raipur	2013-04-28	Delhi Daredevils	Pune Warriors	Pune Warriors	
440	441	2013	Ranchi	2013-05-12	Royal Challengers Bangalore	Kolkata Knight Riders	Kolkata Knight Riders	

	id	season	city	date	team1	team2	toss_winner	toss_deci
457	458	2014	Abu Dhabi	2014-04-16	Kolkata Knight Riders	Mumbai Indians	Kolkata Knight Riders	
458	459	2014	Sharjah	2014-04-17	Delhi Daredevils	Royal Challengers Bangalore	Royal Challengers Bangalore	
461	462	2014	NaN	2014-04-19	Mumbai Indians	Royal Challengers Bangalore	Royal Challengers Bangalore	

```
In [ ]: # Drop duplicate rows based on the 'city' column
# Then return the shape (rows, columns) of the resulting DataFrame

data.drop_duplicates(subset=['city']).shape
```

```
Out[ ]: (31, 18)
```

```
In [ ]: # Drop duplicate rows based on both 'city' and 'season' columns
# Keeps the first occurrence of each unique combination of city and season

data.drop_duplicates(subset=['city', 'season'])
```

Out[]:

	id	season	city	date	team1	team2	toss_winner	toss_deci
0	1	2017	Hyderabad	2017-04-05	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	
1	2	2017	Pune	2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	
2	3	2017	Rajkot	2017-04-07	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders	
3	4	2017	Indore	2017-04-08	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab	
4	5	2017	Bangalore	2017-04-08	Royal Challengers Bangalore	Delhi Daredevils	Royal Challengers Bangalore	
...	...	...	...	...	...	...	...	...
583	584	2016	Hyderabad	2016-04-16	Sunrisers Hyderabad	Kolkata Knight Riders	Sunrisers Hyderabad	
591	592	2016	Pune	2016-04-22	Royal Challengers Bangalore	Rising Pune Supergiants	Rising Pune Supergiants	
612	613	2016	Visakhapatnam	2016-05-08	Sunrisers Hyderabad	Mumbai Indians	Mumbai Indians	
626	627	2016	Kanpur	2016-05-19	Kolkata Knight Riders	Gujarat Lions	Gujarat Lions	
627	628	2016	Raipur	2016-05-20	Sunrisers Hyderabad	Delhi Daredevils	Delhi Daredevils	

107 rows × 18 columns



```
In [ ]: # Drop duplicate rows based on the combination of 'city' and 'season'
# Then get the shape (number of rows and columns) of the resulting DataFrame

data.drop_duplicates(subset=['city', 'season']).shape
```

Out[]: (107, 18)

In []: *# Drop duplicate rows based on 'season', keeping the last occurrence of each season*
Then sort the resulting DataFrame by 'season' in ascending order

```
data.drop_duplicates(subset='season', keep='last').sort_values('season')
```

Out[]:

	id	season	city	date	team1	team2	toss_winner	toss_decision
--	----	--------	------	------	-------	-------	-------------	---------------

116	117	2008	Mumbai	2008-06-01	Chennai Super Kings	Rajasthan Royals	Rajasthan Royals	field
-----	-----	------	--------	------------	---------------------	------------------	------------------	-------

173	174	2009	Johannesburg	2009-05-24	Deccan Chargers	Royal Challengers Bangalore	Royal Challengers Bangalore	field
-----	-----	------	--------------	------------	-----------------	-----------------------------	-----------------------------	-------

233	234	2010	Mumbai	2010-04-25	Chennai Super Kings	Mumbai Indians	Chennai Super Kings	bat
-----	-----	------	--------	------------	---------------------	----------------	---------------------	-----

306	307	2011	Chennai	2011-05-28	Chennai Super Kings	Royal Challengers Bangalore	Chennai Super Kings	bat
-----	-----	------	---------	------------	---------------------	-----------------------------	---------------------	-----

380	381	2012	Chennai	2012-05-27	Chennai Super Kings	Kolkata Knight Riders	Chennai Super Kings	bat
-----	-----	------	---------	------------	---------------------	-----------------------	---------------------	-----

456	457	2013	Kolkata	2013-05-26	Mumbai Indians	Chennai Super Kings	Mumbai Indians	bat
-----	-----	------	---------	------------	----------------	---------------------	----------------	-----

516	517	2014	Bangalore	2014-06-01	Kings XI Punjab	Kolkata Knight Riders	Kolkata Knight Riders	field
-----	-----	------	-----------	------------	-----------------	-----------------------	-----------------------	-------

575	576	2015	Kolkata	2015-05-24	Mumbai Indians	Chennai Super Kings	Chennai Super Kings	field
-----	-----	------	---------	------------	----------------	---------------------	---------------------	-------

635	636	2016	Bangalore	2016-05-29	Sunrisers Hyderabad	Royal Challengers Bangalore	Sunrisers Hyderabad	bat
-----	-----	------	-----------	------------	---------------------	-----------------------------	---------------------	-----

58	59	2017	Hyderabad	2017-05-21	Mumbai Indians	Rising Pune Supergiant	Mumbai Indians	bat
----	----	------	-----------	------------	----------------	------------------------	----------------	-----



In []: *# Drop duplicate rows based on 'season', keeping the last occurrence for each season*
Select only the 'season' and 'winner' columns
Sort the resulting DataFrame by 'season' in ascending order

```
data.drop_duplicates(subset='season', keep='last')[['season', 'winner']].sort_va
```

Out[]:

	season	winner
116	2008	Rajasthan Royals
173	2009	Deccan Chargers
233	2010	Chennai Super Kings
306	2011	Chennai Super Kings
380	2012	Kolkata Knight Riders
456	2013	Mumbai Indians
516	2014	Kolkata Knight Riders
575	2015	Mumbai Indians
635	2016	Sunrisers Hyderabad
58	2017	Mumbai Indians